

Section 5.1 If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies that

$$|f(x) - f(y)| \leq (x-y)^2 \text{ for all } x, y \in \mathbb{R}$$

Then f is constant.

Proof. Since $\left| \frac{f(x) - f(y)}{x-y} \right| \leq |x-y|$ for all $x, y \in \mathbb{R}$,

Given $x \in \mathbb{R}$, $f'(x)$ exists with $f'(x) = 0$.

Now, by the Mean Value Theorem, for any $a < b$, there exists $c \in (a, b)$ s.t.

$$f(a) - f(b) = (a-b) \cdot f'(c) = 0$$

Thus f is constant.

Section 5.2. Let $f: (a, b) \rightarrow \mathbb{R}$ be given. If $f'(x) > 0$ for all $x \in (a, b)$, then f is injective.

Moreover, the inverse $f^{-1}: f[(a, b)] \rightarrow (a, b)$ exists and is differentiable with

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (a < x < b)$$

Proof. Let $a < x < y < b$. Then, by the Mean-Value Theorem, there exists $c \in (x, y)$ s.t.

$$f(y) - f(x) = f'(c) \cdot (y-x) > 0$$

Therefore, f is strictly increasing, thus injective. Now, the restriction

$$f: (a, b) \rightarrow f[(a, b)]$$

is bijective, thus there exists an inverse

$$f^{-1}: f[(a, b)] \rightarrow (a, b): f(x) \mapsto x$$

Now, $(f^{-1} \circ f)(x) = x$ implies that

[Section 5.3. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given with $|g'| \leq M$ for some $M > 0$.

Fix $\varepsilon > 0$, and define $f(x) = x + \varepsilon \cdot g(x)$. prove that f is injective if ε is small enough.

Proof. If $\varepsilon < \frac{1}{M}$, then $f'(x) = 1 + \varepsilon \cdot g'(x) > 0$. Therefore f is injective.